

Causal Inference: Foundations & Estimation

Chapter 11: Why Model?

Part II: Causal Inference with Models — Study Notes & Applied Reference

1 Introduction: The Transition to Model-Based Causal Inference

In Part I of *What If (Causal Inference Without Models)*, all methods—standardization, inverse probability (IP) weighting, stratification, and matching—relied on **nonparametric estimators** under saturated models. Those non-parametric methods allowed data to “speak for themselves” without imposing a priori restrictions on the shape of response curves or conditional distributions.

However, non-parametric estimation requires that every combination of treatment levels A and confounders L contains sufficient observations. As we transition to Part II (*Causal Inference with Models*), we address realistic research scenarios where treatment A is continuous or multi-valued, or covariates L are high-dimensional. Under these conditions, nonparametric estimation breaks down due to empty cells, necessitating parametric models.

Core Takeaway: The Necessity of Parametric Models

When data cannot speak for themselves because of high dimensionality or continuous treatments, parametric models compensate for sparse data by imposing structural restrictions (*smoothing*) that allow us to borrow information across observations.

2 11.1 Data Cannot Speak for Themselves

Consider a study measuring a continuous outcome Y (e.g., CD4 cell count, or customer retention months) across different levels of treatment A :

- **Dichotomous Treatment** ($A \in \{0, 1\}$): With 16 subjects divided into $A = 0$ and $A = 1$, sample averages $\hat{E}[Y|A = 0]$ and $\hat{E}[Y|A = 1]$ are easily calculated and unbiased for population means.
- **Polytomous Treatment** ($A \in \{1, 2, 3, 4\}$): As treatment categories increase, sample sizes per category decrease, increasing the length of 95% confidence intervals.
- **Continuous Treatment** ($A \in [0, 100]$ **mg/day**): When treatment takes many integer or continuous values, many specific doses (e.g., $A = 90$) have **zero observed individuals** in the sample.

For continuous treatments, the sample average $\hat{E}[Y|A = a]$ is mathematically **undefined** for any level a where no data was recorded. Without a model, we cannot estimate the outcome mean for unobserved treatment levels.

3 11.2 Parametric Estimators of the Conditional Mean

To estimate $E[Y|A = 90]$ when no subject received dose 90, we introduce an *a priori* restriction on the shape of the conditional mean function $E[Y|A]$.

3.1 Linear Conditional Mean Model

Assuming $E[Y|A]$ is a straight line across all values of A :

$$E[Y|A] = \theta_0 + \theta_1 A \quad (1)$$

where θ_0 is the intercept and θ_1 is the slope parameter representing the change in $E[Y|A]$ per unit change in treatment A .

Mechanism: Borrowing Information

Ordinary least squares (OLS) uses *all* data points across the entire range of A to estimate $\hat{\theta}_0$ and $\hat{\theta}_1$. The predicted value $\hat{E}[Y|A = 90] = \hat{\theta}_0 + 90\hat{\theta}_1$ “borrows information” from individuals receiving other dose levels ($A \neq 90$).

3.2 Model Definition & Misspecification Risk

A **model** is defined as an *a priori* mathematical restriction on the joint distribution of the data. While parametric models enable estimation where data are absent, their validity hinges on the untestable condition of **no model misspecification**. If the true relationship is curved but modeled linearly, predictions will be systematically biased.

4 11.3 Nonparametric Estimators vs. Saturated Models

Table 1: Comparison of Estimator Types and Assumptions

Property	Nonparametric / Saturated	Parametric / Parsimonious
Data Restriction	None (lets data speak for themselves)	Imposes structural functional forms
Parameters	Equal to number of treatment/strata cells	Fixed small vector $\theta = (\theta_0, \dots, \theta_p)^T$
Fisher Consistency	Yes (Fisher consistent under saturated model)	No (unless true distribution matches model)
High Dimensions	Fails when cells are sparse/empty	Smooths over empty cells / high L
Key Assumption	Identifiability assumptions only	Identifiability + No Model Misspecification

- **Fisher Consistency:** An estimator is Fisher-consistent if, when calculated on the entire population rather than a sample, it yields the exact population parameter without model restrictions.

- **Saturated Model:** A model with as many parameters as possible outcome/treatment combinations (e.g., $E[Y|A] = \theta_0 + \theta_1 A$ for binary A). It imposes zero restrictions and is equivalent to nonparametric estimation.
- **Parsimonious Model:** A model with few parameters used to estimate many population quantities (e.g., 2 parameters for 100 treatment levels).

5 11.4 Smoothing

Parametric modeling can be conceptualized as transforming noisy sample data into smooth response surfaces:

- **Polynomial Models:** Adding quadratic ($\theta_2 A^2$) or cubic ($\theta_3 A^3$) terms creates curved response surfaces ($E[Y|A] = \theta_0 + \theta_1 A + \theta_2 A^2$).
- **Degree of Smoothing:** As the number of parameters increases, the estimated curve becomes more “wiggly” (less smooth). A model with as many parameters as data points interpolates all sample points (zero smoothing).
- **Multivariate Surfaces:** When adjusting for a vector of covariates $L = (L_1, \dots, L_p)$, the model defines a hyperdimensional surface.
- **Local Smoothing / Nonparametric Regression:** Kernel regression and Generalized Additive Models (GAMs) borrow information locally from near neighbors within a bandwidth h , providing flexible smoothing without global functional form restrictions.

6 11.5 The Bias-Variance Trade-Off

Every parametric modeling decision involves an inherent trade-off between bias and variance:

The Fundamental Bias-Variance Trade-Off

- **Parsimonious / Smoother Models (Fewer Parameters):** Higher risk of model misspecification bias, BUT lower sampling variance (narrower confidence intervals).
- **Complex / Less Smooth Models (More Parameters):** Lower risk of model misspecification bias, BUT higher sampling variance (wider confidence intervals).

If an overly simple 2-parameter model is biased due to misspecification, its nominal 95% confidence intervals will fail to cover the true parameter 95% of the time. In practice, investigators perform **sensitivity analyses** across alternative model specifications (e.g., linear vs. polynomial vs. cubic splines) to evaluate how robust causal effect estimates are to modeling choices.